

Generative AI for 2D Character Animation

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Figure 1: Text-to-image outputs from our Dreambooth SDXL character models for FRED, VERA and MS. MENACE.

ABSTRACT

In this pilot project, we teamed up with artists to develop new workflows for 2D animation while producing a short educational cartoon. We created several workflows streamlining the animation process, bringing the artists’ vision to the screen more effectively.

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1 INTRODUCTION

We produced a short educational cartoon using novel generative AI workflows. Educational videos are under-served relative to entertainment videos, and educators don’t generally have the skill or talent to produce animations that teach real concepts. We adopted a 2D style that can be more forgiving of the kind of geometric inconsistencies often produced by generative AI, which limit the perceived quality of realistic or 3D-style imagery. Our narrative occurs in a fantasy world with some non-human characters, giving further flexibility for variable character appearance.

2 WORKFLOWS

Previous generative animation projects [Maas et al. 2024] have focused on automating story simulation and visual design, but in this work, our goal is to accelerate the process of character animation. We identified several promising workflows (see Figure 2), bringing artists’ vision to the screen more effectively.

SDXL fine tuning. All our workflows begin with fine-tuned diffusion models. In early stages of production with a small crew, we had only a handful of illustrations for each character, so we fine-tuned

SDXL [Podell et al. 2023] using DreamBooth [Ruiz et al. 2023]. Since full custom fine-tuning for each character is costly, we explored training LoRAs [Hu et al. 2021], but the results did not reliably reproduce the style or design of our characters. Even fully fine-tuned models sometimes produced inconsistent proportions, line weights, or paint colors. Non-human characteristics — like a tail or three eyes — were especially problematic, often blending with other body parts. For example, notice FRED’s tail and MS. MENACE’s eyes in Figure 1. In most cases we overcame these inconsistencies with artist sketches as additional conditioning, using ControlNet [Zhang et al. 2023]. Our characters are illustrated in a flat 2D style, but surprisingly we found it useful to apply depth conditioning in addition to sketch conditioning. To produce masks for compositing, each character’s images were composited onto a contrasting background color before training. In this way the models tended to output a constant background color, which could be easily chroma-keyed using Adobe After Effects or Nuke [Ado 2024a; Nuk 2024].

We tried introducing motion to these characters using several image-to-video models. However, the results generally included extraneous motions that did not match the sharp, minimal, highly-focused motion style of 2D animation we desired. Instead, we identified three general workflows that we found most successful, with AI portions implemented in ComfyUI [comfyanonymous 2024].

AI puppets. For simple animations, in which a character stays close to a single primary key pose, we used an artist sketch of that pose as input to ControlNet to generate a single inked-and-painted hero frame. Then SegmentAnything [Kirillov et al. 2023] separated that frame into body parts. The resulting layers were imported into Adobe Character Animator [Ado 2024b] for pose animation

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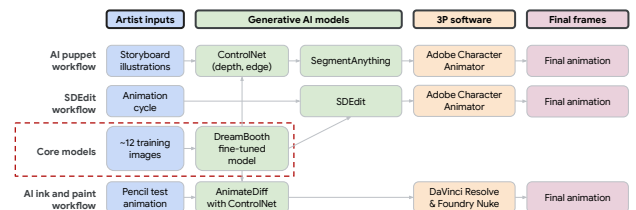


Figure 2: Three of our generative AI animation workflows.

and audio lip-sync. We removed the generated mouth shape using Photoshop [Pho 2024], and replaced it with a library of hand-drawn shapes with Character Animator’s lip-sync tools.

AI ink-and-paint. For shots with more dramatic pose variations, animators drew pencil artwork per-frame, and we used AnimateDiff [Guo et al. 2024] with ControlNet to generate consistent line and coloring (see Figure 3). To ensure the lines conformed to the inputs, we use Canny-edge or sketch ControlNets, as well as depth ControlNet used in the puppet workflow. In some cases, we iterated this process, feeding the output animation back into ControlNet as an additional input, finding this reduced the incidence of incorrectly colored regions and “off-model” character poses. Color deviations and edge noise in the output were corrected using DaVinci Resolve and Nuke [DaV 2024; Nuk 2024].

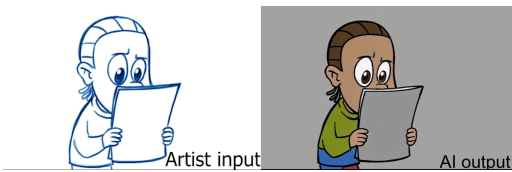


Figure 3: A sample frame of input and output for the character LUTHER from our AI ink and paint workflow.

SDEdit. For our radioactive FIREFLY, who is supposed to be a little crazy, we filtered frames from an input smooth animation cycle using SDEdit [Meng et al. 2022] with our SDXL fine-tuned model. Each output frame was similar to the input frame, but not identical (see Figure 4). This randomness provided the FIREFLY a sense of constant, erratic motion.

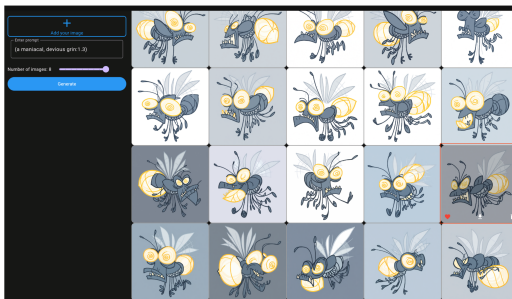


Figure 4: Sample generations from the FIREFLY model. We took advantage of the randomness of diffusion generations to amplify this character’s erratic and unpredictable nature!

3 CONCLUSION

AI workflows enabled animators to focus more time and effort on story, design and characterization, and less on tedious tasks such as ink-and-paint. However, substantial manual cleanup was required to reach our target quality bar, and over the course of production we fine-tuned our models repeatedly with different inputs to gain better intuitions. The ink-and-paint workflow needed very clean input

lines to work well; cleaner than our animators might normally use for a pencil test. Nonetheless, diffusion models exceeded our initial expectations, enabling us to complete about 50 shots with only one full-time and two part-time animators over a period of approximately eight weeks. We expect results to improve using base models trained or fine-tuned with more cartoon and animation samples, and with new conditioning strategies to constrain elements of character design such as proportion, line weight, and color palette. Visit <https://genai-2d-character-animation.github.io/> for additional information and results.

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